



Spin-Orbit Coupled Calculations

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Outline

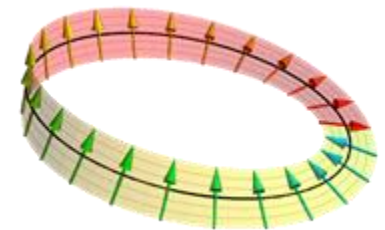
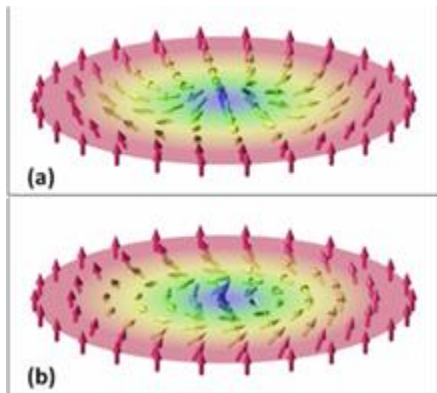
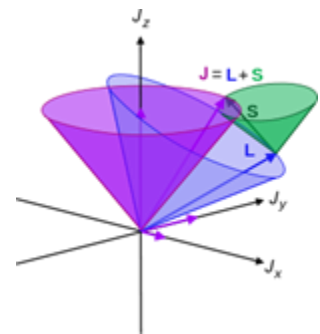


1. Relativistic Quantum Mechanics:

- a. Schrödinger Theory*
- b. Dirac Theory*
- c. Pauli Theory*

2. Relativistic Effects in the CRYSTAL Code

- a. The Pseudopotential Approximation*
- b. The Two-Component Self-Consistent Field*



Outline

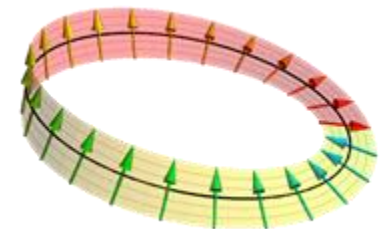
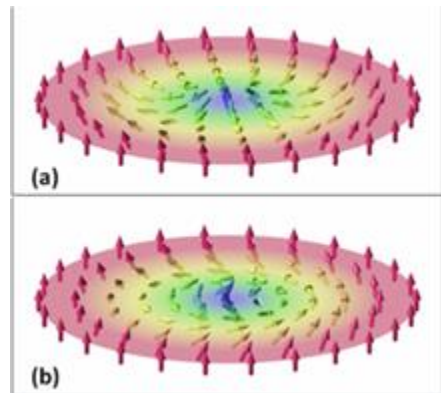
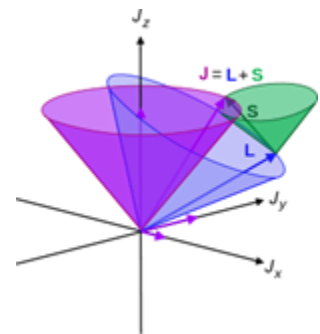


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57 La Lanthanum 3	58 Ce Cerium 3,4	59 Pr Praseodymium 3	60 Nd Neodymium 3	61 Pm Promethium 3	62 Sm Samarium 3	63 Eu Europium 2,3	64 Gd Gadolinium 3	65 Tb Terbium 3	66 Dy Dysprosium 3	67 Ho Holmium 3	68 Er Erbium 3	69 Tm Thulium 3	70 Yb Ytterbium 3	71 Lu Lutetium 3
89 Ac Actinium 3	90 Th Thorium 4	91 Pa Protactinium 5	92 U Uranium 6	93 Np Neptunium 5	94 Pu Plutonium 4	95 Am Americium 3	96 Cm Curium 3	97 Bk Berkelium 3	98 Cf Californium 3	99 Es Einsteinium 3	100 Fm Fermium 3	101 Md Mendelevium 3	102 No Nobelium 2	103 Lr Lawrencium 3

Relativistic Quantum Mechanics

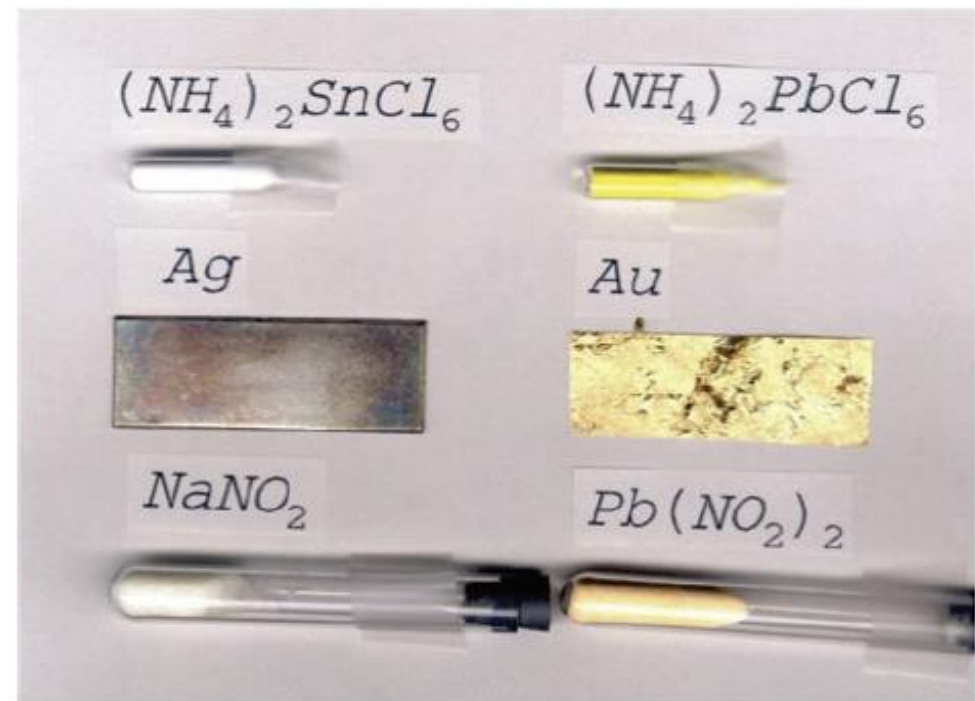
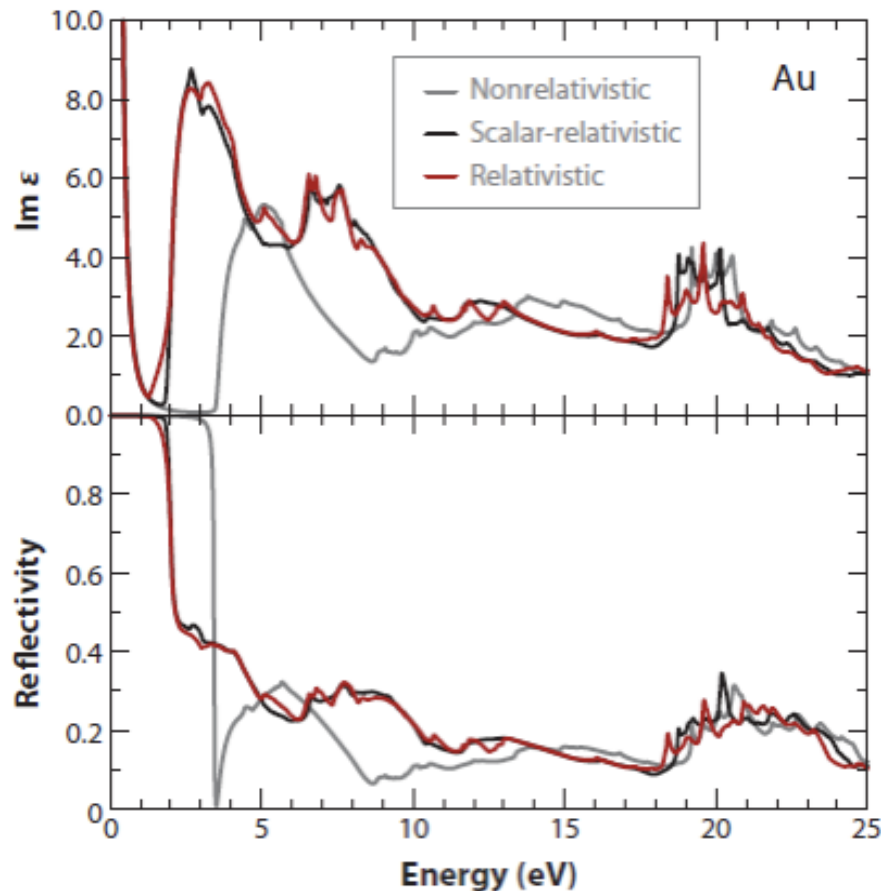


1 H Hydrogen -1,1 2 He Helium		1 H Hydrogen 1.008 s p d f 7s 6s 5s 4s 3s 2s 1s 7p 6p 5p 4p 3p 2p 6d 5d 4d 3d 5f 4f $\ell = 0$ $m = 0$ $n = 1$										5 B Boron 3 6 C Carbon -4,4 7 N Nitrogen -3,3,5 8 O Oxygen -2 9 F Fluorine -1 10 Ne Neon						11 Na Sodium 1 12 Mg Magnesium 2 13 Al Aluminum 3 14 Si Silicon -4,4 15 P Phosphorus -3,3,5 16 S Sulfur -2,2,4,6 17 Cl Chlorine -1,1,3,5,7 18 Ar Argon						19 K Potassium 1 20 Ca Calcium 2 21 Sc Scandium 3 22 Ti Titanium 4 23 V Vanadium 5 24 Cr Chromium 3,6 25 Mn Manganese 2,4,7 26 Fe Iron 2,3 27 Co Cobalt 2,3 28 Ni Nickel 2 29 Cu Copper 2 30 Zn Zinc 2 31 Ga Gallium 3 32 Ge Germanium -4,2,4 33 As Arsenic -3,3,5 34 Se Selenium -2,2,4,6 35 Br Bromine -1,1,3,5,7 36 Kr Krypton 2						37 Rb Rubidium 1 38 Sr Strontium 2 39 Y Yttrium 3 40 Zr Zirconium 4 41 Nb Niobium 5 42 Mo Molybdenum 4,6 43 Tc Technetium 4,7 44 Ru Ruthenium 3,4 45 Rh Rhodium 3 46 Pd Palladium 2,4 47 Ag Silver 1 48 Cd Cadmium 2 49 In Indium 3 50 Sn Tin -4,2,4 51 Sb Antimony -3,3,5 52 Te Tellurium -2,2,4,6 53 I Iodine -1,1,3,5,7 54 Xe Xenon 2,4,6						55 Cs Caesium 1 56 Ba Barium 2 57-71 Lanthanides 72 Hf Hafnium 4 73 Ta Tantalum 5 74 W Tungsten 4,6 75 Re Rhenium 4 76 Os Osmium 4 77 Ir Iridium 3,4 78 Pt Platinum 2,4 79 Au Gold 3 80 Hg Mercury 1,2 81 Tl Thallium 1,3 82 Pb Lead 2,4 83 Bi Bismuth 3 84 Po Polonium -2,2,4 85 At Astatine -1,1 86 Rn Radon 2						87 Fr Francium 1 88 Ra Radium 2 89-103 Actinides 104 Rf Rutherfordium 4 105 Db Dubnium 5 106 Sg Seaborgium 6 107 Bh Bohrium 7 108 Hs Hassium 8 109 Mt Meitnerium 110 Ds Darmstadtium 111 Rg Roentgenium 112 Cn Copernicium 113 Nh Nihonium 114 Fl Flerovium 115 Mc Moscovium 116 Lv Livermorium 117 Ts Tennessine 118 Og Oganesson						57 La Lanthanum 3 58 Ce Cerium 3,4 59 Pr Praseodymium 3 60 Nd Neodymium 3 61 Pm Promethium 3 62 Sm Samarium 3 63 Eu Europium 2,3 64 Gd Gadolinium 3 65 Tb Terbium 3 66 Dy Dysprosium 3 67 Ho Holmium 3 68 Er Erbium 3 69 Tm Thulium 3 70 Yb Ytterbium 3 71 Lu Lutetium 3						89 Ac Actinium 3 90 Th Thorium 4 91 Pa Protactinium 5 92 U Uranium 6 93 Np Neptunium 5 94 Pu Plutonium 4 95 Am Americium 3 96 Cm Curium 3 97 Bk Berkelium 3 98 Cf Californium 3 99 Es Einsteinium 3 100 Fm Fermium 3 101 Md Mendelevium 3 102 No Nobelium 2 103 Lr Lawrencium 3					
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Common oxidation states are shown in bold beneath the element closeup.

Periodic Table Design & Interface Copyright © 1997 Michael Dayah Ptable.com Last updated Jun 16, 2017

Relativistic Quantum Mechanics

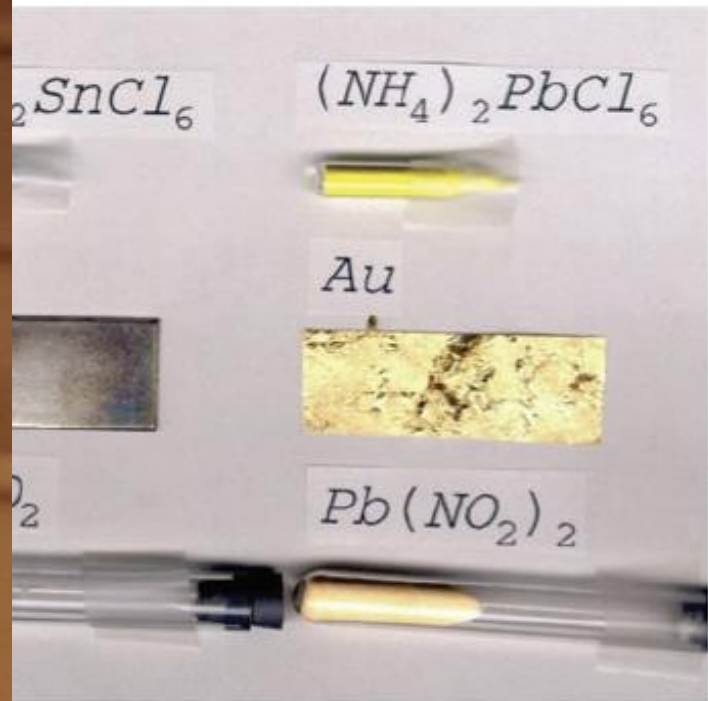
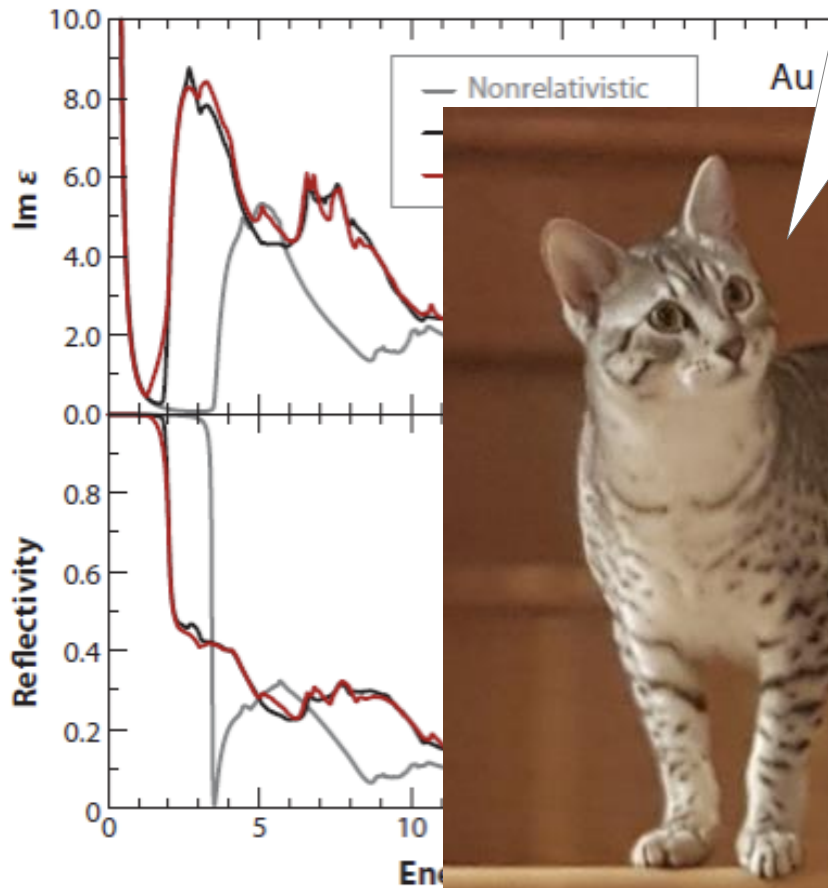


Gold would have a silver colour without relativistic effects!!

Pyykkö 2012



Wow!!!!
So interesting!



Gold would have a silver colour without relativistic effects!!

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Schrödinger Theory



We start with the Classical energy expression for a free electron:

$$E = \frac{\mathbf{p}^2}{2m}$$

and apply the Quantum Mechanical Correspondance Principle
(classical observables $\rightarrow$ QM operators acting on the wave function):

$$E \rightarrow i\hbar \frac{\partial}{\partial t} \qquad \mathbf{p} \rightarrow -i\hbar \nabla$$

to get:

$$\frac{-\hbar^2}{2m} \nabla^2 \Psi(\mathbf{r}, t) = i\hbar \frac{\partial}{\partial t} \Psi(\mathbf{r}, t)$$



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Dirac Theory



Take the energy-momentum relation of special relativity:

$$E = c \sqrt{\mathbf{p}^2 + m^2 c^2}$$



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by inspection we get:

$$\boldsymbol{\alpha}_i = \begin{pmatrix} \mathbf{0}_2 & \boldsymbol{\sigma}_i \\ \boldsymbol{\sigma}_i & \mathbf{0}_2 \end{pmatrix} \quad \boldsymbol{\beta} = \begin{pmatrix} \mathbb{I}_2 & \mathbf{0}_2 \\ \mathbf{0}_2 & -\mathbb{I}_2 \end{pmatrix}$$

$$\boldsymbol{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \boldsymbol{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \boldsymbol{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



Dirac Theory



$$\left(-i\hbar c\boldsymbol{\alpha} \cdot \boldsymbol{\nabla} + \beta mc^2\right) \boldsymbol{\Psi}(\mathbf{r}, t) = i\hbar \frac{\partial}{\partial t} \boldsymbol{\Psi}(\mathbf{r}, t)$$

- First spatial and time derivatives
- Complex Hamiltonian
- Four component matrices $\rightarrow$ four component wave function



Dirac Theory: the Dirac Wave Function



$$E = \pm \sqrt{\mathbf{p}^2 c^2 + m^2 c^4} \rightarrow E \in (-\infty, -mc^2] \cup [mc^2, \infty)$$

$$\psi(\mathbf{r}) = \begin{pmatrix} \psi^{L,\alpha}(\mathbf{r}) \\ \psi^{L,\beta}(\mathbf{r}) \\ \psi^{S,\alpha}(\mathbf{r}) \\ \psi^{S,\beta}(\mathbf{r}) \end{pmatrix}$$

➡ Mostly (spin up or down) Electrons

➡ Mostly (spin up or down) Positrons

Pauli Theory



We start with the time-independent Dirac Equation:

$$\left(c\boldsymbol{\alpha} \cdot \mathbf{p} + \beta' mc^2 + \mathbb{I}_4 \otimes V \right) \psi = E\psi$$

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and we introduce the following shorthand notation for the large (L) and small (S) components :

$$\boldsymbol{\psi}^L = \left[\psi^{L,\alpha}, \psi^{L,\beta} \right]^T \text{ and } \boldsymbol{\psi}^S = \left[\psi^{S,\alpha}, \psi^{S,\beta} \right]^T$$

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Then, taking into account the form of the Dirac matrices, we obtain:

$$(V - E) \boldsymbol{\psi}^L + c (\boldsymbol{\sigma} \cdot \mathbf{p}) \boldsymbol{\psi}^S = \mathbf{0}_2$$

$$c (\boldsymbol{\sigma} \cdot \mathbf{p}) \boldsymbol{\psi}^L + (V - E - 2mc^2) \boldsymbol{\psi}^S = \mathbf{0}_2$$

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Isolating for the S component in the second Eq. gives:

$$c\psi^S = \frac{1}{2m} \left[1 - \frac{V - E}{2mc^2} \right]^{-1} (\boldsymbol{\sigma} \cdot \mathbf{p}) \psi^L$$

Pauli Theory



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Let us now use the geometric series expression:

$$\left[1 - \frac{V - E}{2mc^2} \right]^{-1} = 1 + \frac{V - E}{2mc^2} + \left[\frac{V - E}{2mc^2} \right]^2 + \dots$$

Pauli Theory



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Keeping the first two terms of the geometric series and substitution back into the Dirac Eq. For the small component gives (after renormalization):

$$\mathbf{H}_{Pauli} \tilde{\psi}^L = E \tilde{\psi}^L$$

Pauli Theory



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where:

$$\mathbf{H}_{Pauli} = \mathbb{I}_2 \otimes \left[\frac{\mathbf{p}^2}{2m} + V - \frac{1}{8m^3 c^2} \mathbf{p}^4 + \frac{1}{8m^2 c^2} \nabla^2 V \right] + \frac{1}{4m^2 c^2} \boldsymbol{\sigma} \cdot (\nabla V) \times \mathbf{p}$$

Pauli Theory



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non relativistic



Pauli Theory



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If V is the electron-nuclear potential:

$$\mathbf{h}_{SO}(i) = \frac{e^2}{4m^2c^2} \sum_A \frac{Z_A}{r_{Ai}^3} \mathbf{L}_{Ai} \cdot \boldsymbol{\sigma},$$

$\mathbf{L}$ is the electron-nuclear angular momentum:

$$\mathbf{L}_{Ai} = \mathbf{r}_{Ai} \times \mathbf{p}_i$$

The Pauli equation is for one electron. V is not the many-electron potential!

SOC leads to:

→ splitting of otherwise degenerate p , d , f levels

→ change of symmetry of the wavefunction



Outline

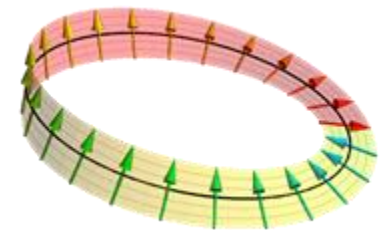
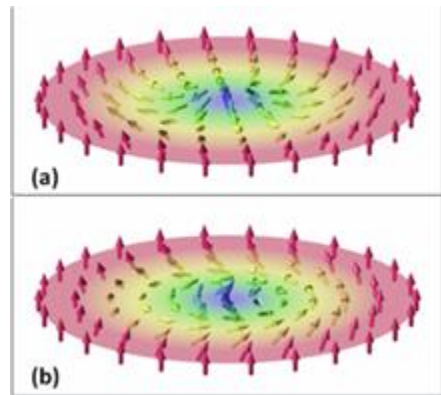
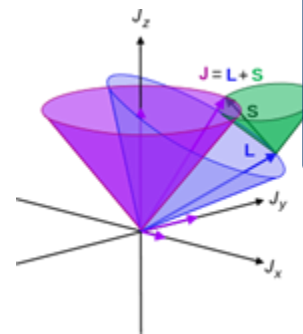


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Pseudopotential Approximation



- Core orbitals make the calculation more expensive but often do not participate “directly” in the physics or chemistry
- Relativistic effects (SR and SOC) have a large “direct” effect on the core orbitals but the effect on the valence is “indirect” (although, still important, contrary to what Dirac initially thought!)

Pseudopotential Approximation



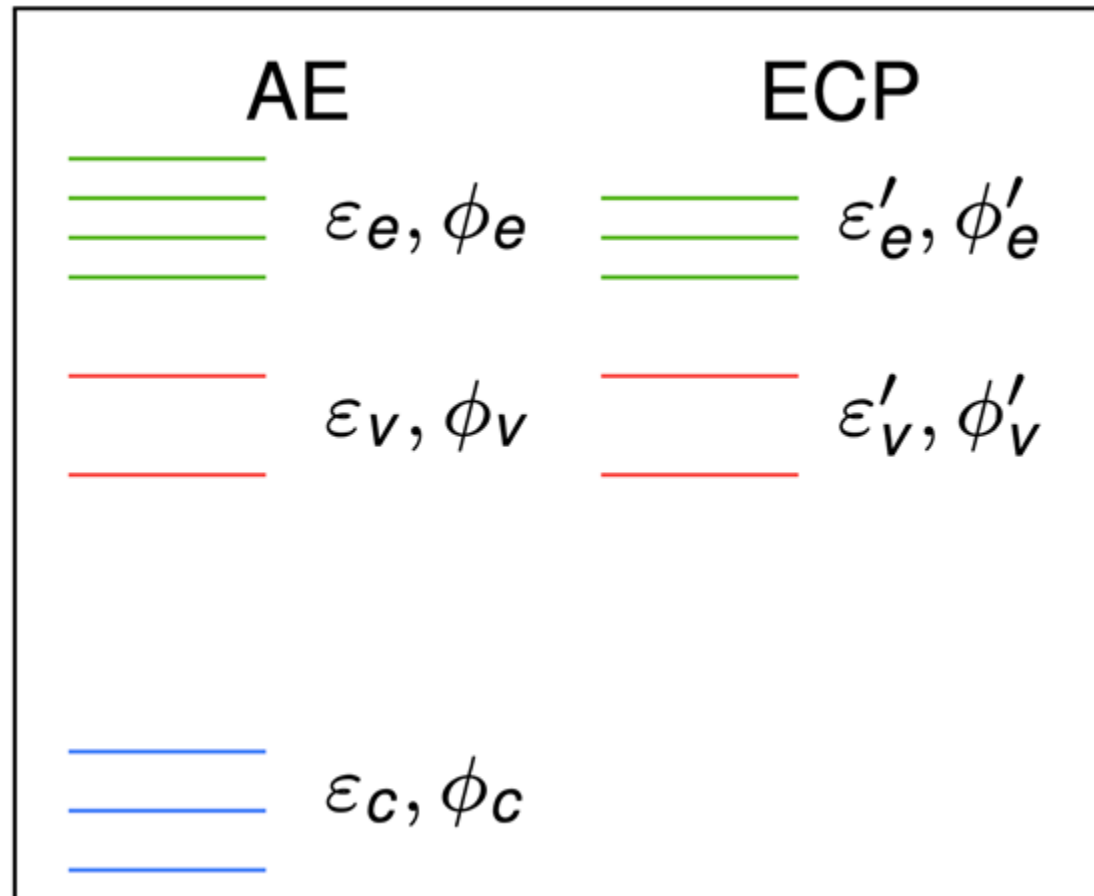
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- Dirac (1929):

The general theory of quantum mechanics is now almost complete, the imperfections that still remain being in connection with the exact fitting in of the theory with relativity ideas. These give rise to difficulties only when high-speed particles are involved, and are therefore of no importance in the consideration of atomic and molecular structure and ordinary chemical reactions, in which it is, indeed, usually sufficiently accurate if one neglects relativity variation of mass with velocity and assumes only Coulomb forces between the various electrons and atomic nuclei. The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble. It there-

Pseudopotential Approximation



- Represent all core orbitals by an effective potential that “screens” the valence
- Do a four-component accurate relativistic wave function calculation on individual atom(s) for the effective potential(s) that replaces the core
- After the effective potential has been obtained, solve a “valence-only” equation
- ECP: Effective Core Potential

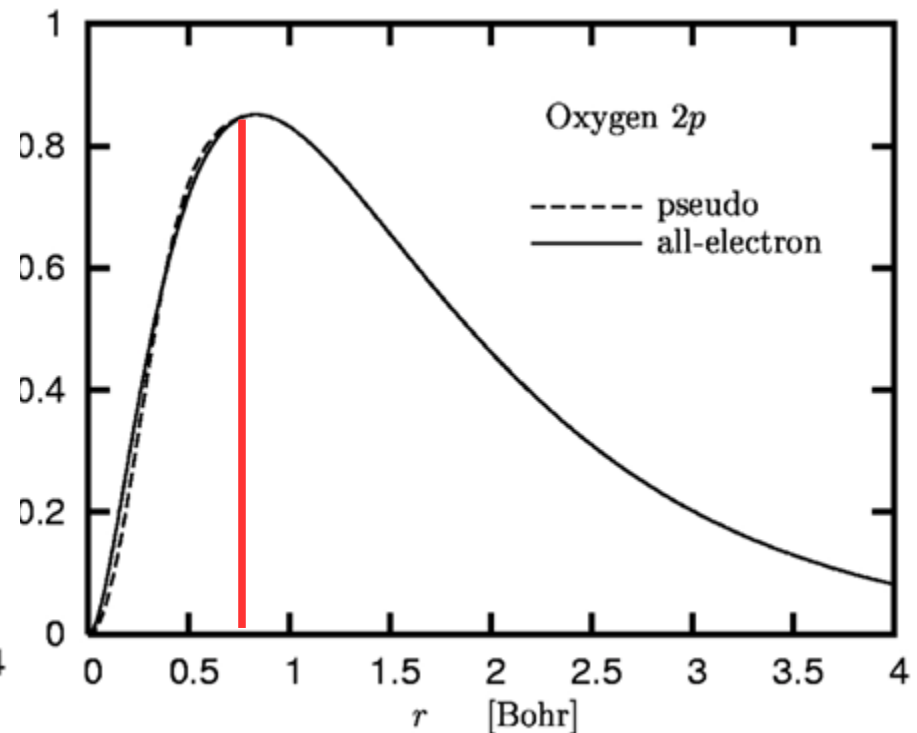
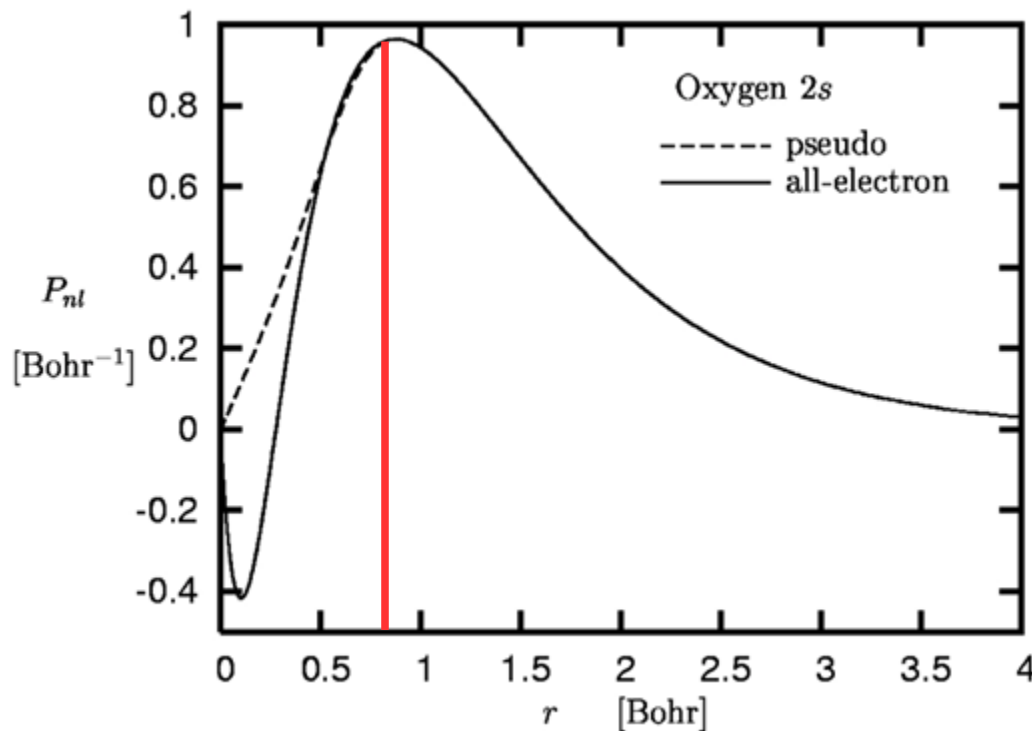


Pseudopotential Approximation



- In the "shape-consistent" method, the core electrons are replaced by an effective potential which is fitted in such a way that the pseudo orbitals reproduce the exact orbitals in the valence region

$$U_{lj}^{REP}(r_{Ai}) = \sum_k c_{klj}^A r_{Ai}^{n_{klj}} e^{-a_{klj}^A r_{Ai}^2}$$



Pseudopotential Approximation



We can rigorously derive the ideas of the pseudopotential approach through the Philips-Kleinman equation.[Philips and Kleinman, 1959]

The starting point is the single-particle equation for the parent all-electron (AE) model (e.g. Hartree-Fock, KS-DFT, Dirac-Hartree-Fock, etc...):[Dolg and Cao, 2012]

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$$\hat{H}_{\text{eff}}|\varphi_a\rangle = \varepsilon_a|\varphi_a\rangle, a \in \{v, c\} \quad \langle\varphi_a|\varphi_{a'}\rangle = \delta_{aa'}, a, a' \in \{v, c\}$$

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where the orbitals are subject to orthonormality.

We can now define an arbitrary normalized function

(which represents the “pseudovalence” orbitals – the orbitals in the pseudopotential calculation) as:

$$|\varphi_p\rangle = N_p(|\varphi_v\rangle + \sum_c a_c|\varphi_c\rangle)$$

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$$|\varphi_p\rangle = N_p(|\varphi_v\rangle + \sum_c a_c|\varphi_c\rangle)$$

The mixing coefficients are determined by the orthonormality condition. Then, solving for valence orbital and inserting the result back into the single-particle equation gives:

$$(\hat{H}_{\text{eff}} + \hat{V}^{\text{PK}})|\varphi_p\rangle = \varepsilon_v|\varphi_p\rangle$$

where:

$$\hat{V}^{\text{PK}} = \sum_c (\varepsilon_v - \varepsilon_c)|\varphi_c\rangle\langle\varphi_c|$$

Pseudopotential Approximation



We have all of the ideas of the pseudopotential approximation from the Philips-Kleinman equation:

$$(\hat{H}_{\text{eff}} + \hat{V}^{\text{PK}})|\varphi_p\rangle = \varepsilon_v|\varphi_p\rangle$$

1. $\hat{H}_{\text{eff}}$ is a valence-only Hamiltonian
2. $\hat{V}^{\text{PK}}$ is an effective potential that replaces the core orbitals (comprising of linear coefficients and projection operators onto core states)
3. the pseudopotential approximation allows (in principle) to preserve the true eigenvalues ε_v , even though the pseudovalence orbitals $|\varphi_p\rangle$ are different from the valence orbitals $|\varphi_v\rangle$

Pseudopotential Approximation



Explicit form of the pseudopotential in the two-component case:

$$W_{ps}^{\text{REP}} = C + \sum_{l=0}^{\infty} \sum_{j=j_{\alpha}}^{j_{\beta}} U_{lj}^{\text{REP}} P_{lj}^{\text{Pauli}}$$

$$W_{ps}^{\text{AREP}} = \sum_{l=0}^{\infty} U_l^{\text{AREP}} P_l$$

$$W_{ps}^{\text{SOREP}} = \sum_{l=0}^{\infty} U_l^{\text{SOREP}} P_l \vec{L} \cdot \vec{\sigma} P_l$$

$$U_l^{\text{AREP}} = \frac{1}{2l+1} \left[l U_{lj^{\beta}}^{\text{REP}} + (l+1) U_{lj^{\alpha}}^{\text{REP}} \right] \equiv \sum_{k=1}^{M_{\ell}} r^{n_{k\ell}} C_{k\ell}^{\text{AREP}} e^{-\alpha_{k\ell} r^2}$$

$$U_l^{\text{SOREP}} = \frac{2}{2l+1} \left[U_{lj^{\alpha}}^{\text{REP}} - U_{lj^{\beta}}^{\text{REP}} \right] \equiv \frac{2}{2l+1} \sum_{k=1}^{M_{\ell}} r^{n_{k\ell}} C_{k\ell}^{\text{SOREP}} e^{-\alpha_{k\ell} r^2}$$

Pseudopotential Approximation



Explicit form of the pseudopotential in the two-component case:

$$W_{ps}^{\text{SOREP}} = \sum_a \sum_{l=0}^{\infty} U_l^{\text{SOREP}} P_l L_a \sigma^a P_l = \sum_{\mu, a} \sigma^a A_{\mu}^a p_{\mu}$$

$$A_{\mu}^a = \sum_{\nu} \sum_{l=0}^{\infty} U_l^{\text{SOREP}} P_l \varepsilon_{\mu}^a{}_{\nu} r_{\nu} P_l$$

$$P_l = \sum_{m=-l}^l |l, m\rangle \langle l, m|$$

$$\int d^3r \langle \Psi | l, m \rangle = \int dr r^2 \int d\Omega \Psi^*(\mathbf{r}) X_m^l(\Omega)$$

Pseudopotential Approximation



1 H Hydrogen																	2 He Helium																		
3 Li Lithium	4 Be Beryllium																	5 B Boron	6 C Carbon	7 N Nitrogen	8 O Oxygen	9 F Fluorine	10 Ne Neon												
11 Na Sodium	12 Mg Magnesium																	13 Al Aluminium	14 Si Silicon	15 P Phosphorus	16 S Sulfur	17 Cl Chlorine	18 Ar Argon												
19 K Potassium	20 Ca Calcium	21 Sc Scandium	22 Ti Titanium	23 V Vanadium	24 Cr Chromium	25 Mn Manganese	26 Fe Iron	27 Co Cobalt	28 Ni Nickel	29 Cu Copper	30 Zn Zinc	31 Ga Gallium	32 Ge Germanium	33 As Arsenic	34 Se Selenium	35 Br Bromine	36 Kr Krypton	37 Rb Rubidium	38 Sr Strontium	39 Y Yttrium	40 Zr Zirconium	41 Nb Niobium	42 Mo Molybdenum	43 Tc Technetium	44 Ru Ruthenium	45 Rh Rhodium	46 Pd Palladium	47 Ag Silver	48 Cd Cadmium	49 In Indium	50 Sn Tin	51 Sb Antimony	52 Te Tellurium	53 I Iodine	54 Xe Xenon
55 Cs Cesium	56 Ba Barium	57 La Lanthanum	72 Hf Hafnium	73 Ta Tantalum	74 W Tungsten	75 Re Rhenium	76 Os Osmium	77 Ir Iridium	78 Pt Platinum	79 Au Gold	80 Hg Mercury	81 Tl Thallium	82 Pb Lead	83 Bi Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon	87 Fr Francium	88 Ra Radium	89 Ac Actinium	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium	113 Nh Nihonium	114 Fl Flerovium	115 Mc Moscovium	116 Lv Livermorium	117 Ts Tennessine	118 Og Oganesson

Columbus SC

Columbus LC

Stuttgart SC

Stuttgart LC

Columbus SH

Stuttgart SH

* 58 Ce Cerium	59 Pr Praseodymium	60 Nd Neodymium	61 Pm Promethium	62 Sm Samarium	63 Eu Europium	64 Gd Gadolinium	65 Tb Terbium	66 Dy Dysprosium	67 Ho Holmium	68 Er Erbium	69 Tm Thulium	70 Yb Ytterbium	71 Lu Lutetium
** 90 Th Thorium	91 Pa Protactinium	92 U Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

Two-Component Self-Consistent Field Approach



As was anticipated for the Pauli theory, the standard UHF procedure:

$$\begin{aligned}\mathbf{F}_{\{\mathbf{k}_m\}}^{\alpha\alpha} \mathbf{c}_{\{\mathbf{k}_m\}}^{\alpha\alpha} &= \mathbf{S}_{\{\mathbf{k}_m\}}^{\alpha\alpha} \mathbf{c}_{\{\mathbf{k}_m\}}^{\alpha\alpha} \epsilon_{\{\mathbf{k}_m\}}^{\alpha\alpha} \\ \mathbf{F}_{\{\mathbf{k}_m\}}^{\beta\beta} \mathbf{c}_{\{\mathbf{k}_m\}}^{\beta\beta} &= \mathbf{S}_{\{\mathbf{k}_m\}}^{\beta\beta} \mathbf{c}_{\{\mathbf{k}_m\}}^{\beta\beta} \epsilon_{\{\mathbf{k}_m\}}^{\beta\beta}\end{aligned}$$

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with two uncoupled Eqs. for each spin, must now be modified to account for the fact that each orbital can have a spin-up or spin-down component. We saw that the Hamiltonian is now a 2 X 2 matrix, and the wave function (a Pauli spinor) has a 2 X 1 structure, so we get:

$$\begin{bmatrix} \mathbf{F}_{\{\mathbf{k}_i\}}^{\alpha\alpha} & \mathbf{F}_{\{\mathbf{k}_i\}}^{\alpha\beta} \\ \mathbf{F}_{\{\mathbf{k}_i\}}^{\beta\alpha} & \mathbf{F}_{\{\mathbf{k}_i\}}^{\beta\beta} \end{bmatrix} \begin{bmatrix} \mathbf{c}_{\{\mathbf{k}_i\}}^{\alpha} \\ \mathbf{c}_{\{\mathbf{k}_i\}}^{\beta} \end{bmatrix} = \begin{bmatrix} \mathbf{S}_{\{\mathbf{k}_i\}}^{\alpha\alpha} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_{\{\mathbf{k}_i\}}^{\beta\beta} \end{bmatrix} \begin{bmatrix} \mathbf{c}_{\{\mathbf{k}_i\}}^{\alpha} \\ \mathbf{c}_{\{\mathbf{k}_i\}}^{\beta} \end{bmatrix} \epsilon_{\{\mathbf{k}_i\}}$$

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The Fock matrix in momentum-space is constructed via the Fourier relation:

$$F_{\mu\nu\{\mathbf{k}_i\}}^{\sigma\sigma'} = \sum_{\mathbf{g}} e^{i\mathbf{k}_i \cdot \mathbf{g}} F_{\mu\nu\{\mathbf{g}\}}^{\sigma\sigma'}$$

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and the Fock matrix in position space reads:

$$F_{\mu\nu\{\mathbf{g}\}}^{\sigma\sigma'} = \int d\mathbf{r}_j \chi_{\mu}(\mathbf{r}_j - \mathbf{A}_{\mu}) F^{\sigma\sigma'} \chi_{\nu}(\mathbf{r}_j - \mathbf{A}_{\nu} - \mathbf{g})$$

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